



Alberta Public Health Disease Management Guidelines

Carbapenemase-Producing Organisms (CPOs)

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Carbapenemase-Producing Organisms (CPO) | Primary and Preventative Health Services, Government of Alberta

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Table of Contents

Case Definition	5
Confirmed.....	5
Reporting Requirements	6
Laboratories	6
Primary Care Alberta and Indigenous Services Canada (ISC)	6
Clinical Assessment and Epidemiology.....	7
Etiology	7
Clinical Presentation.....	7
Diagnosis	7
Treatment.....	8
Transmission	8
Incubation Period	8
Period of Communicability.....	8
Host Susceptibility	8
Incidence.....	8
Public Health Management.....	10
Key Investigation.....	10
Management of a Case.....	10
Management of Contacts	10
Preventative Measures.....	10

Revision History

Revision Date	Document Section	Description of Revision
July 2025	Case Definition	Updated per ProvLab review.
	Reporting	- Removed “Physicians, health practitioners and others.” Under section 15 of the Alberta Public Health Act, only laboratories and the MOH have mandatory reporting requirements. - Added minor edits to provide clarity.
	Clinical Assessment and Epidemiology	New section added.
	Public Health Management	New section added.

Case Definition

Confirmed

Laboratory confirmation of a carbapenemase-producing organism by:

- Positive genotypic (e.g., PCR) or secondary phenotypic (e.g., mCIM, lateral flow assay or other) testing in any *Enterobacterales*^(A) isolate that is carbapenem non-susceptible^(B);

OR

- Positive genotypic testing (e.g., PCR) in an *Acinetobacter* spp. isolate that is carbapenem non-susceptible^(B).

^(A) Enterobacterales order includes, but is not limited to:

- Common: Species of *Escherichia coli*, *Klebsiella*, *Citrobacter*, *Enterobacter*, *Hafnia*, *Morganella*, *Proteus*, *Providencia*, *Serratia*.
- Uncommon: Species of *Edwardsiella*, *Erwinia*, *Kluyvera*, *Salmonella*, *Shigella*, *Yersinia*
- Rare: *Raoultella* species, *Pantoea* species

^(B) Non-susceptible includes intermediate and resistant to carbapenem antibiotics (e.g., meropenem, imipenem or ertapenem) per CLSI M100 clinical breakpoints for carbapenems.

Reporting Requirements

Laboratories

All laboratories shall report all positive laboratory results confirming a carbapenemase in any carbapenem non-susceptible *Enterobacterales* or *Acinetobacter* isolate (refer to case definition) by mail, fax or electronic submission within 48 hours (two business days) to the:

- Chief Medical Officer of Health (CMOH) (or designate), and
- Zone MOH (or designate).

Primary Care Alberta and Indigenous Services Canada (ISC)

- The Zone MOH (or designate) where the case currently resides shall forward the [Carbapenemase-Producing Organisms \(CPO\) Report](#) form for all confirmed cases to the CMOH (or designate) within two weeks of notification and any amendments within four weeks.
- For out-of-province and out-of-country cases, the MOH (or designate) should forward the following information to the CMOH (or designate) by phone, fax or electronic submission within 48 hours (two business days) of initial notification:
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - positive laboratory report, and
 - other relevant clinical/epidemiological information.

Clinical Assessment and Epidemiology

Etiology

Carbapenemase-producing organisms (CPOs) are a subgroup of carbapenem-resistant organisms (CROs) (See Table 1). Other subgroups of CROs include carbapenem-resistant *Enterobacterales* (CREs) and carbapenemase-producing *Enterobacterales* (CPEs).

Table 1: Relationship between CROs, CPOs, CREs and CPEs^(1,2)

Organism Group	How They Resist Antibiotics	CROs	CPOs	CREs	CPEs
Lactose non-fermenters Examples: <i>Acinetobacter baumannii</i> , <i>Pseudomonas aeruginosa</i> and <i>Stenotrophomonas maltophilia</i>	Natural (built-in) resistance	✓	x	x	x
	Carbapenemase is acquired ^(c)	✓	✓	x	x
Enterobacterales Examples: <i>Klebsiella pneumoniae</i> and <i>Escherichia coli</i>	ESBL or AmpC plus porin loss	✓	x	✓	x
	Carbapenemase is acquired	✓	✓	✓	✓

CPOs are specific strains of bacteria such as *Klebsiella*, *Escherichia coli* (*E. coli*), *Acinetobacter*, and *Pseudomonas* that are resistant to antibiotics known as carbapenems and other β -lactam antibiotics, including penicillin and cephalosporins because they produce an enzyme called a carbapenemase.⁽³⁾ CPOs are concerning because of their ability to transfer resistance easily across different genera and species of bacteria. The six most frequently identified carbapenemases are divided into 3 classes:⁽⁴⁾

- **Class A:** *K. pneumoniae* carbapenemase (KPC),
- **Class B Metallo- β -lactamases (MBLs):** New Delhi MBL (NDM), Verona Integron-encoded MBL (VIM), and imipenemase MBL (IMP); and
- **Class D:** Oxacillinase-48 (OXA-48) and OXA-48-like.

Clinical Presentation

Individuals colonized with CPOs are asymptomatic; however, CPOs can cause infections if they gain access to sterile body sites (e.g., lungs, bladder, bloodstream).⁽⁵⁾ CPOs may cause difficult-to-treat infections in any part of the body, including cystitis, pyelonephritis, septicemia, pneumonia, peritonitis, wound infections and meningitis.⁽⁶⁾ Mortality in patients with CPO bacteremia may be up to 50%.

Diagnosis

CPO colonization is identified through screening protocols established by Infection Prevention and Control (IPC) and culture-based identification methods established by Alberta microbiology laboratories.

In Alberta, the diagnosis of an infection caused by a CPO requires culture of an appropriate clinical specimen and isolation of a carbapenem non-susceptible *Enterobacterales* or *Acinetobacter* species. Carbapenemase production is confirmed by a combination of phenotypic (identifies if organism produces carbapenemase) and genotypic (detects specific resistance genes) methods.

^(c) Resistance genes entered bacteria through mobile genetic elements (e.g., plasmids).

Treatment

Treatment decisions for patients with CPO infections are made by healthcare providers on a case-by-case basis. Treatment is not usually required for people who are colonized with CPOs.⁽⁷⁾ Consultation with an infectious disease specialist is recommended.

Transmission

CPOs are most often transmitted via person-to-person contact with the wounds or stool of an infected or colonized individual or contact with contaminated surfaces.⁽⁷⁾ CPOs are not spread through the air.

In healthcare settings, CPOs are transmitted from person-to-person, often via the hands of health care personnel or through contaminated medical equipment. Additionally, sink drains and toilets are increasingly recognized as an environmental reservoir and CPO transmission source.^(8–10)

CPOs can also be transmitted to and from companion animals.⁽¹¹⁾

Incubation Period

Unknown. Symptomatic CPO infection can occur weeks to years after being initially colonized, especially in individuals who are immunocompromised (See Host Susceptibility).^(12,13)

Period of Communicability

The period of communicability may potentially be as long as the organism is present in the body, typically in the gastrointestinal tract.⁽⁸⁾ A person can test positive intermittently on repeat screening and may be colonized for months to over a year.⁽¹⁴⁾

Host Susceptibility

Individuals who have had surgery, dialysis, or been admitted to health care facilities outside of Canada are at increased risk of acquiring the bacteria and developing infection.⁽⁷⁾

Factors that impair the function of the immune system (e.g., hematologic malignancy), and interventions which permit colonizing bacteria to invade (e.g., indwelling devices) may also increase the probability of infection.^(15,16)

Incidence

There is no consensus on the definition of an endemic CPO country.^(D) Generally, it refers to widespread and persistent presence of specific types of CPOs that are 1) commonly found in healthcare settings and potentially in the community and 2) there is high incidence of infection or colonization. Detection of this complex group of organisms (and surveillance for them) require considerable healthcare resources (including high functioning microbiology laboratories) that are not available in many parts of the world. The global spread of CPOs continues to evolve, driven by international travel and shifts in healthcare practices.⁽⁴⁾

NDMs are the most prevalent carbapenemase among carbapenem-resistant *E. coli* (CR-EC), found in 75 countries.^(17,18) It is endemic in South Asia (India, Pakistan, Bangladesh) and parts of Africa; also common in Egypt, Thailand, and Vietnam.^(18,19)

^(D) There is very little data on global distribution of CPOs as a whole. Many surveillance programs focus on specific subsets of CPOs. CR-EC data is often used to represent the global distribution of CPOs because *E. coli* is a common and clinically significant species among *Enterobacterales*, making it a good indicator for carbapenem resistance trends.

OXAs are the second most common carbapenemase in CR-EC globally.⁽¹⁸⁾ OXA-48-like enzymes are endemic in North Africa, the Middle East, and parts of Europe.⁽²⁰⁾

KPCs are the third most common carbapenemase in CR-EC globally.⁽¹⁸⁾ It is endemic in the United States, China, and parts of Southern Europe (Italy and Greece).^(18,21)

Healthcare-associated antimicrobial resistance, including carbapenemase-producing *Enterobacterales* (CPE) infections and colonization, are monitored locally by IPC and at the national level by the [Canadian Antimicrobial Resistance Surveillance System \(CARSS\)](#).

CPOs have been notifiable in Alberta since 2013. Most cases reported in Alberta are due to NDM-producing bacteria.

For more information on Alberta CPO case numbers and incidence rates, see the [Interactive Health Data Application \(IHDA\)](#).

Public Health Management

Key Investigation

- Confirm that the individual meets case definition.
- Obtain history of illness including whether the individual is colonized or has symptomatic infection, and date of symptom onset if symptomatic.
- Determine if the individual has any of the following exposures within the previous 12 months:
 - travel outside Alberta, including any hospitalizations,
 - recent immigration to Alberta, including any hospitalizations prior to arrival,
 - hospitalizations or healthcare services received in Alberta,
 - epidemiological link (i.e., close contact) to a confirmed CPO case or outbreak. This includes stay in a facility/unit with known, active CPO transmission, or
 - household contact of someone who travelled.

Management of a Case

- In the healthcare setting (e.g., acute care, congregate care), infection prevention and control (IPC) should be consulted regarding isolation precautions.
 - For information on infection prevention and control precautions refer to the following: [IPC Resource Manuals](#).
- For cases (including colonized cases) at home, the following information may be provided: [MyHealthAlberta: CPO – What to do at home](#).
- Cases should also be encouraged to avoid other secondary preventable infections by ensuring their immunization status is up to date. Refer to the [Alberta Immunization Policy \(AIP\)](#).

Management of Contacts

- There is no public health management of contacts.

Preventative Measures

Health care staff:

- To prevent the spread of CPOs, health care personnel and facilities should refer to the [IPC Resource Manuals for more information](#).

General public:

- To reduce the risk of CPO transmission, the public should be aware of the following:^(7,22)
 - Perform hand hygiene regularly by washing hands with soap and water when visibly soiled; otherwise, use an alcohol-based hand sanitizer.
 - Take antibiotics only as prescribed by a health care provider. Do not take leftover antibiotics or obtain them without a prescription, as this can contribute to antimicrobial resistance.
 - Medical procedures and treatments outside of Canada can be associated with increased risk of acquiring antimicrobial resistant organisms, including CPOs.
 - When seeking treatment or undergoing a procedure (e.g. emergency department visit, elective surgery, dialysis), be sure to advise your healthcare provider if you have had medical procedures or treatments performed outside of Canada in the previous 12 months.

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